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Scoping review of artificial intelligence approaches for solar energy forecasting addressing methodological developments data practices and operational implications

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Abstract

Accurate solar energy forecasting has become increasingly critical with the growing integration of solar power into modern energy systems. In response, artificial intelligence (AI)–based techniques have been widely employed to address the nonlinear, nonstationary, and weather-driven characteristics of solar generation. This scoping review systematically maps the literature on AI-based solar energy forecasting with emphasis on modeling approaches, data practices, and operational implications. A comprehensive search was conducted across Scopus, Web of Science, IEEE Xplore, ScienceDirect, and Google Scholar to identify eligible studies and extract information on model categories, input data characteristics, forecasting horizons, and application contexts. The synthesized evidence reveals a clear temporal evolution from classical machine learning toward deep learning, hybrid ensembles, and transformer-based architectures, with solar irradiance forecasting and short-term to day-ahead horizons dominating the research landscape. Ensemble and hybrid models consistently demonstrated improved robustness under variable atmospheric conditions, whereas transformer-based approaches showed potential for multi-step prediction tasks. However, most studies relied on site-specific datasets with limited cross-regional validation, highlighting persistent challenges in model transferability. Importantly, forecasting accuracy was closely linked to operational objectives, including grid stability, reserve management, and market participation, yet deployment-related considerations such as uncertainty quantification, retraining strategies, and explainability were rarely evaluated. This review identifies a key evolutionary transition in solar forecasting research, characterized by the progression from classical machine learning approaches toward deep learning, hybrid, and transformer-based architectures.

Keywords Artificial intelligence, Machine learning, Photovoltaic power forecasting, Solar energy forecasting, Solar irradiance

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1 Introduction

Solar energy is one of the most abundant renewable energy resources, playing a central role in the global transition toward sustainable energy systems. Climate change, greenhouse gas emissions, and the depletion of conventional fossil fuel reserves have accelerated the adoption of solar energy as a clean and reliable alternative within the global energy mix [1]. Consequently, countries worldwide are investing substantial capital in the expansion of photovoltaic (PV) infrastructure to meet rising energy demands while simultaneously reducing carbon footprints [2]. However, solar energy generation is inherently dependent on multiple environmental, atmospheric, and operational variables, including solar irradiance, cloud cover, temperature, humidity, and seasonal variations [3]. This variability poses significant challenges to energy scheduling and reliable power output, particularly under high solar penetration scenarios [4]. As a result, accurate solar energy forecasting has become essential for optimal operation of solar power systems.

Traditionally, solar energy forecasting has relied on statistical and mathematical models, such as regression-based approaches and time-series techniques [5–7]. However, such approaches struggle to capture the nonlinear interactions among the multifaceted predictors influencing solar irradiance and PV power generation [8, 9]. These limitations lead to inconsistencies and reduced forecasting accuracy, which proportionally affect grid operations and energy storage planning [10]. With the rapid expansion of renewable energy systems and the increasing demand for real-time operational decision-making, there is a pressing need for high-accuracy forecasting methodologies capable of handling diverse data sources and dynamic environmental conditions.

Artificial intelligence (AI) has emerged as a powerful and transformative tool for solar energy forecasting. Particularly, machine learning (ML) approaches have gained particular prominence due to their ability to process large volumes of historical and real-time meteorological data [11–13]. In this context, ML refers to computational methods that automatically learn patterns from data to make predictions without relying on predefined physical equations, while deep learning represents an advanced subset of ML that uses multi-layer neural networks to extract complex temporal or spatial relationships from large datasets. As a result, ML-driven solar forecasting provides a potential to significantly optimize energy dispatch and improve overall prediction efficiency.

Despite the growing body of literature demonstrating the application of ML in solar energy forecasting, the field remains methodologically fragmented. Existing studies differ substantially in forecasting targets, prediction horizons, data sources, evaluation metrics, and reporting practices, making it challenging to obtain an integrated understanding of methodological developments and their operational relevance. Such fragmentation limits reproducibility, complicates comparative benchmarking across studies, and hinders the identification of forecasting approaches that are transferable to diverse climatic and operational environments. In addition, the literature has experienced a notable temporal shift from classical ML techniques toward deep learning, hybrid ensemble models, and more recently transformer-based architectures, reflecting rapid methodological evolution in the field. Transformer-based models use attention mechanisms to identify relationships across time steps and multiple variables, which has shown promise for improving multi-step and day-ahead forecasting performance. Moreover, recent advancements, including hybrid architectures, transformer-based models,

and uncertainty-aware learning approaches, have emerged rapidly, creating the need for a consolidated and up-to-date synthesis. Therefore, a comprehensive review is required to systematically map the evolving landscape of AI-based solar energy forecasting and identify research gaps that may hinder practical deployment. The present scoping review aims to (i) categorize forecasting models and architectures, (ii) examine variations in data sources, feature engineering strategies, and evaluation metrics, and (iii) analyze operational implications of forecasting approaches while identifying methodological gaps and promising strategies for operational deployment in real-world energy systems.

2 Methodology

2.1 Review design and methodological framework

The present review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines. The methodological framework was deployed as per Arksey and O'Malley, with refinements suggested by Levac and colleagues [14–16]. The main objective of this review was to comprehensively map the application of ML techniques in solar energy forecasting and to identify methodological trends, performance patterns, and research gaps. Specifically, the review sought to address; how different ML paradigms have been applied to solar irradiance and PV power forecasting; how forecasting horizons and data sources influence methodological choices; what performance metrics are most commonly employed to evaluate model accuracy, and which emerging approaches show promise for operational deployment.

2.2 Information sources

Literature search was undertaken utilizing databases such as Scopus, Web of Science Core Collection, IEEE Xplore, and ScienceDirect. In addition, Google Scholar was used as a supplementary source to identify recently published or in-press articles that may not yet have been indexed in traditional databases. These databases were selected to ensure broad coverage of both methodological and application-oriented research.

2.3 Search strategy

Core search terms were formulated that focussed on forecasting targets, such as solar irradiance and photovoltaic power generation, and were combined with terms describing ML and AI methodologies. Boolean operators were used to integrate terms such as "solar energy forecasting," "solar irradiance prediction," and "PV power forecasting" with "machine learning," "deep learning," "artificial intelligence," "ensemble learning," and "transformer models." Searches were conducted independently in each database using syntax adapted to platform-specific indexing rules. Database-specific search strategies employed for the identification of studies is presented in Table 1.

2.4 Eligibility criteria

Studies were included if they were peer-reviewed original research articles or conference papers published between January 2015 and November 2025. Eligible studies were required to apply at least one ML, deep learning, ensemble, emerging, federated learning, physics-informed approaches, or hybrid modeling approach to the forecasting of solar irradiance or PV power output. Studies were excluded if; (i) they relied exclusively

Table 1 Database specific keyword search strings

Database	Search fields used	Exact search string
Scopus	TITLE-ABS-KEY	TITLE-ABS-KEY (("solar energy forecasting" OR "solar irradiance forecasting" OR "solar irradiance prediction" OR "PV power forecasting" OR "photovoltaic power prediction") AND ("machine learning" OR "deep learning" OR "artificial intelligence" OR "ensemble learning" OR "hybrid model*" OR "transformer model*"))
Web of Science Core Collection	Topic (TS)	TS = (("solar energy forecasting" OR "solar irradiance forecasting" OR "solar irradiance prediction" OR "photovoltaic power forecasting" OR "PV power prediction") AND ("machine learning" OR "deep learning" OR "artificial intelligence" OR "ensemble learning" OR "hybrid model*" OR "transformer*"))
IEEE Xplore	Metadata (Title, Abstract, Index Terms)	("solar energy forecasting" OR "solar irradiance prediction" OR "PV power forecasting") AND ("machine learning" OR "deep learning" OR "artificial intelligence" OR "neural network*" OR "ensemble learning" OR "transformer")
ScienceDirect	Title, Abstract, Keywords	TITLE-ABSTR-KEY (("solar energy forecasting" OR "solar irradiance prediction" OR "photovoltaic power forecasting") AND ("machine learning" OR "deep learning" OR "artificial intelligence" OR "ensemble learning" OR "hybrid model*" OR "transformer model*"))
Google scholar	Full-text search	"solar energy forecasting" OR "solar irradiance prediction" OR "PV power forecasting" AND "machine learning" OR "deep learning" OR "artificial intelligence"

on classical statistical or physical modeling approaches without incorporating ML components, (ii) if they were editorials, tutorials, opinion pieces, or non-peer-reviewed reports, or (iii) if they were published in languages other than English.

2.5 Study selection process

All records retrieved from the database searches were imported into a Zotero (Corporation for Digital Scholarship) and duplicate entries were removed prior to screening. Study selection was conducted in two sequential stages. In the first stage, titles and abstracts were screened to assess relevance to the review objectives. In the second stage, full-text articles were evaluated against the predefined eligibility criteria. Screening was performed independently by two authors to reduce selection bias. Any discrepancies or disagreements were resolved through discussion and consensus. The overall study identification, screening, and inclusion process is presented in PRISMA-ScR flow diagram (Fig. 1).

2.6 Data synthesis and analytical approach

A narrative and thematic synthesis approach was adopted for data analysis. Studies were grouped according to model taxonomy, forecasting horizon, data characteristics, and evaluation methodology. Conceptual mappings were used to highlight patterns in methodological choices, performance trade-offs, and emerging research directions. No primary experimental data were generated or analyzed as part of this study.

3 Results

3.1 Study selection

The database searches identified 5752 records across all sources. After removal of 3606 duplicate records, 2146 unique records remained and were screened based on titles and abstracts. Of these, 1964 records were excluded at the title and abstract screening stage due to irrelevance to solar forecasting applications or absence of ML–based approaches.

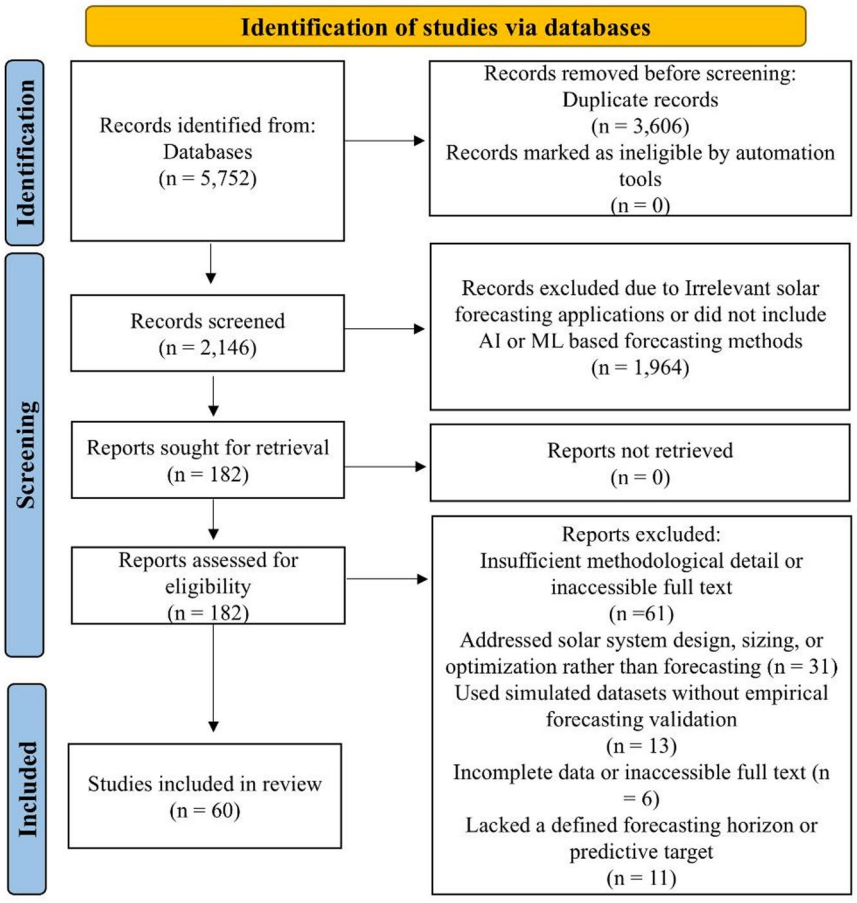


Fig. 1 PRISMA-ScR study selection flow diagram

Full-text assessment was conducted for 182 articles, of which 122 studies were excluded for not meeting the predefined inclusion criteria. Ultimately, 60 studies [17–76] were included in the final scoping review. The study identification, screening, eligibility assessment, and inclusion process is illustrated in the PRISMA-ScR flow diagram (Fig. 1).

3.2 Study characteristics

The included studies display an overall upward trend in publication frequency over time. Early studies (2015–2018) primarily employed classical ML approaches, while a substantial increase in publications was observed from 2019 onward. The geographic distribution of the included studies encompasses Europe, Asia, Africa, the Middle East, and North America, as well as several multi-regional and global-scale investigations. A smaller number of studies adopted multi-site or cross-regional frameworks using satellite or hybrid datasets.

Solar irradiance was the most commonly reported target, particularly in studies focusing on atmospheric modeling, satellite-based prediction, and regional-scale assessments. PV power output forecasting was also widely represented, especially in studies oriented toward operational applications, microgrids, and electricity market participation. The forecasting horizons examined across the studies varied substantially. Ultra-short-term forecasting was commonly reported in studies targeting real-time control and grid stability. Short-term forecasting constituted the largest share of the literature. Day-ahead

forecasting was frequently investigated in the context of scheduling and market-related applications.

Regarding data sources, the majority of studies relied on ground-based meteorological and PV system measurements. Satellite-derived data were used in regional and large-area irradiance forecasting studies, particularly those incorporating remote sensing or sky imagery. Numerical weather prediction (NWP) data were employed in several studies to support short-term and day-ahead forecasts. Additionally, a notable proportion of studies adopted hybrid data approaches, combining ground-based, satellite, and NWP inputs. A comparative summary of representative studies across different modeling paradigms, data sources, and forecasting horizons is presented in Table 2.

3.3 Factors influencing solar power generation

Solar power generation is governed by the availability of solar radiation reaching the Earth's surface, which varies continuously across spatial and temporal scales. Solar radiation arrives at the Earth in the form of solar irradiance, whose magnitude fluctuates across diurnal cycles and seasonal periods [25, 30, 32]. These variations form the fundamental basis for solar energy availability. Environmental and atmospheric conditions further regulate the amount of solar radiation incident at a given location. Atmospheric cloud presence plays a dominant role by attenuating incoming solar radiation through scattering and absorption processes, leading to pronounced short-term variability in surface irradiance [27, 28]. Partial shading causes uneven irradiance distribution across modules, leading to bypass diode activation and the formation of multiple local peaks in the P–V characteristic curve. Such non-linear operating behavior increases forecasting complexity and necessitates intelligent modeling approaches to accurately predict power output under dynamic environmental conditions (Fig. 2) [28, 37]. In addition to cloud effects, air quality parameters such as, aerosols and particulate matter influence irradiance transmission by reducing atmospheric transparency. Aerosol loading contributes to both direct attenuation and diffuse scattering of sunlight, thereby diminishing the effective solar radiation available for energy conversion and reducing PV system efficiency [29, 59]. These effects are particularly relevant in regions experiencing elevated pollution levels or frequent atmospheric haze.

Meteorological parameters such as ambient temperature, relative humidity, wind speed, and wind direction interact with radiative processes. Temperature affects irradiance conversion performance, while wind conditions influence thermal regulation and heat dissipation from PV modules [25, 32, 45]. Humidity and atmospheric pressure modify radiative transfer characteristics and are commonly incorporated alongside irradiance variables in forecasting frameworks [32, 49]. Geographical factors impose additional constraints on solar power generation. Latitude determines the angle of solar incidence and seasonal day-length variation, producing systematic differences in irradiance availability between equatorial and higher-latitude regions [30]. Elevation and site-specific geographic characteristics influence atmospheric thickness, cloud formation tendencies, and radiative losses. Seasonal variations further modulate solar availability, with longer daylight hours and higher solar elevation angles during summer months enhancing generation potential, while winter periods are characterized by reduced irradiance and shorter photoperiods [32]. For PV systems, technological and operational parameters introduce further sources of variability. Panel orientation, tilt

Table 2 Comparative overview of representative studies showcasing artificial intelligence approaches applied in solar irradiance and photovoltaic power forecasting across different data environments and forecasting horizons

Author (year)	Forecast target	Model category	Data source	Forecast horizon	Key findings	Ref No.
Sebastianelli et al. (2024)	Solar irradiance	Machine learning	Satellite meteorological	Short-term	Demonstrated spatial generalization using satellite inputs.	[17]
Amiri et al. (2021)	Solar irradiance	Hybrid evolutionary NN	Ground measurements	Short-term	Evolutionary optimization improved prediction accuracy.	[21]
Heng et al. (2022)	Solar radiation	ANN	Meteorological	Short-term	Backpropagation variants affected convergence and accuracy.	[22]
Obiora et al. (2021)	Solar irradiance	XGBoost	Meteorological	Short-term	Gradient boosting achieved competitive performance.	[23]
Bellido-Jiménez et al. (2021)	Solar radiation	ML regression	Temperature-based	Intra-day	Demonstrated importance of temperature features.	[24]
Buturache & Stancu (2021)	PV power	RNN LSTM GRU	Historical PV and weather	Short-term	Temporal models captured sequential dependencies effectively.	[25]
Gao et al. (2020)	Solar irradiance	CNN–LSTM hybrid	Ground data	Hourly	Spatial–temporal hybrid improved stability.	[26]
Nematchoua et al. (2022)	Solar radiation	ML	European datasets	Daily	Regional forecasting with meteorological inputs.	[29]
]Banik & Biswas (2023)	PV power	RF–CatBoost ensemble	Ground measurements	Short-term	Ensemble improved robustness under variability.	[30]
Sherozbek et al. (2023)	PV power	Transformer	Multivariate weather	Hourly	Attention improved long-horizon performance.	[31]
Pospíchal et al. (2022)	Solar irradiance	Transformer	Meteorological	Day-ahead	Improved temporal dependency modeling.	[32]
Sanchez-Lopez et al. (2024)	Solar irradiance	Decision-tree ensemble	Ground data	Very short-term	Ensemble improved ramp event prediction.	[34]
Mohamed et al. (2022)	Solar forecasting	ML with feature engineering	Microgrid data	Short-term	Feature engineering enhanced predictive capability.	[35]

Table 2 (continued)

Author (year)	Forecast target	Model category	Data source	Forecast horizon	Key findings	Ref No.
Kazem et al. (2022)	PV power	FRNN–PCA hybrid	Meteorological	Long-term	Dimensionality reduction improved performance.	[37]
Garcia-Cuesta et al. (2023)	Solar forecasting	ML dimensionality reduction	Meteorological	Short-term	Feature selection improved model efficiency.	[38]
Sugunaraj & Ranganathan (2024)	Power forecasting	Autoencoder DL	Power system data	Short-term	Autoencoders improved feature extraction.	[40]
Jebli et al. (2021)	Solar energy	ML correlation-based	Meteorological	Short-term	Correlation-guided features improved accuracy.	[42]
Campos et al. (2024)	PV power	LSTM	Historical PV	Short-term	LSTM outperformed classical baselines.	[43]
Asiedu et al. (2024)	PV power	Hybrid ML	Meteorological + PV	Multi-horizon	Hybrid approaches improved stability.	[44]
Visser et al. (2024)	Solar power	Probabilistic ML	Market and weather	Day-ahead	Probabilistic forecasts improved risk-aware planning.	[45]
Renkema et al. (2024)	PV power	Conformal prediction	Meteorological	Short-term	Uncertainty quantification improved reliability.	[48]
Gupta & Singh (2023)	PV power	Multivariate LSTM	Meteorological	Day-ahead	Multivariate inputs improved accuracy.	[49]
Lyu & Eftekharnejad (2024)	Solar generation	Probabilistic DL	Weather datasets	Short-term	Improved prediction under rapid weather changes.	[50]
Taha et al. (2025)	Solar irradiance	ML	Meteorological	Short and long-term	Demonstrated scalability across horizons.	[53]
Hammond et al. (2024)	Solar irradiance	ML	Limited data environments	Short-term	Addressed data scarcity challenges.	[54]
Pereira et al. (2024)	DNI forecasting	ANN + NWP	NWP data	Day-ahead	NWP integration improved forecasting accuracy.	[55]
Takamatsu et al. (2025)	Solar irradiance	Metamodel optimization	Meteorological	Day-ahead	Optimization enhanced prediction efficiency.	[56]

Table 2 (continued)

Author (year)	Forecast target	Model category	Data source	Forecast horizon	Key findings	Ref No.
Almarzooqi et al. (2024)	Solar irradiance	Hybrid ML	Meteorological	Short-term	Hybridization improved nonlinear modeling.	[59]
Vanlalchhuanawmi et al. (2025)	Solar radiation	Multi-model ML DL	Meteorological	Short-term	Multi-model approach improved robustness.	[60]

ANN Artificial Neural Network, *AutoML* Automated Machine Learning, *CNN* Convolutional Neural Network, *DL* Deep Learning, *DNI* Direct Normal Irradiance, *FRNN* Fuzzy Recurrent Neural Network, *GRU* Gated Recurrent Unit, *LSTM* Long Short-Term Memory, *ML* Machine Learning, *NWP* Numerical Weather Prediction, *PCA* Principal Component Analysis, *PV* Photovoltaic, *RF* Random Forest, *RNN* Recurrent Neural Network, *RL* Reinforcement Learning

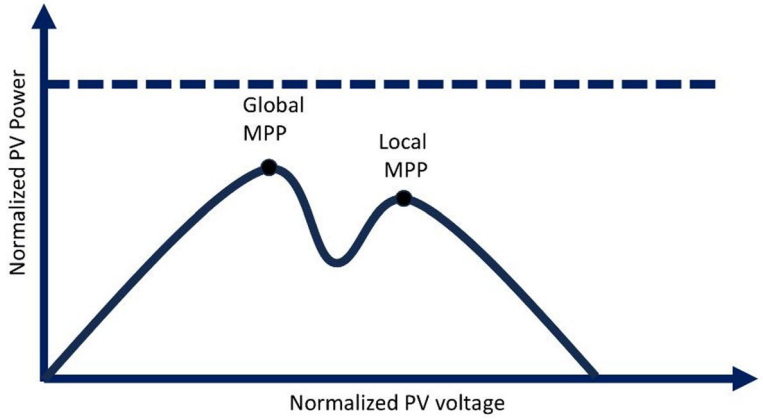


Fig. 2 Schematic representation of photovoltaic power–voltage characteristics under partial shading conditions showing global and local maximum power points. Partial shading produces non-uniform irradiance across modules, creating multiple local maxima in the P–V curve due to mismatch effects and bypass diode activation. This non-linear behavior increases forecasting complexity and highlights the importance of intelligent modeling approaches for accurate photovoltaic power prediction

angle, and tracking configuration govern the effectiveness of solar radiation capture [31]. System-specific characteristics such as inverter performance, module efficiency, shading effects, and degradation over time influence the conversion of irradiance into electrical power [62, 63]. Historical PV power output and operational conditions therefore serve as important descriptors of system behavior under real-world operating environments.

3.4 Classification of forecasting tasks

Solar energy forecasting tasks are broadly classified according to the forecasting target and the temporal horizon over which predictions are generated (Fig. 3). Solar irradiance forecasting focuses on estimating the incoming solar radiation at the Earth's surface and is used for regional-scale solar resource assessment [19, 26, 33]. Such forecasts are relevant in applications such as, atmospheric modeling, satellite-based monitoring, and irradiance-driven performance analysis [20, 51, 52]. In contrast, PV power forecasting directly targets electrical power output from solar installations and incorporates system-level characteristics alongside environmental inputs [38, 40, 45, 54].

The temporal horizon of forecasting further differentiates these tasks. Ultra-short-term forecasting addresses rapid variability in solar output caused primarily by cloud

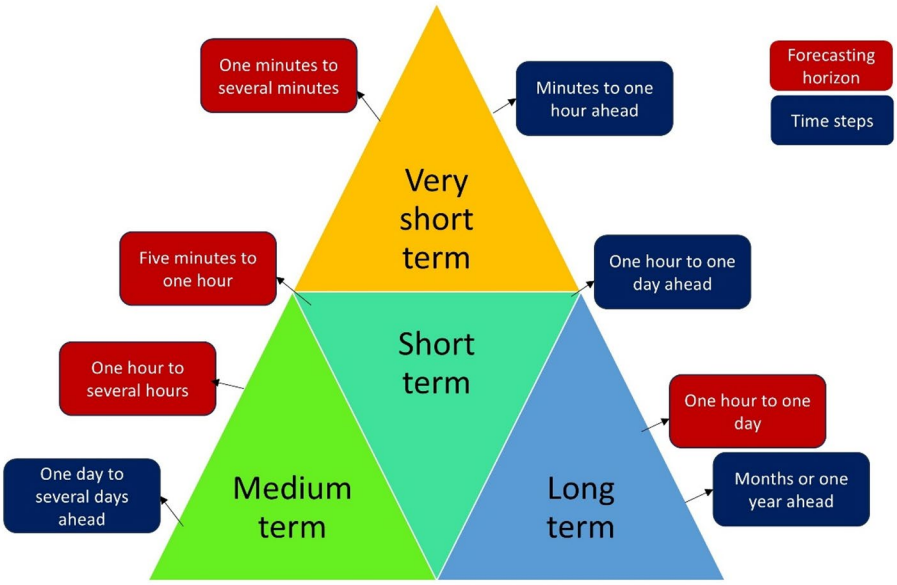


Fig. 3 Solar energy forecasting categories

movement and transient atmospheric conditions. This horizon is closely linked to real-time grid control, ramp event mitigation and power quality management [28, 37, 55]. Image-based methods are commonly associated with this forecasting scale [46, 55]. Short-term forecasting represents the most frequently addressed horizon in solar forecasting applications. This time scale supports operational scheduling load balancing and relies on combinations of historical measurements and near-term meteorological inputs [32, 44, 49]. Likewise, day-ahead forecasting plays a critical role in energy market operations and grid scheduling. Predictions at this horizon perform bidding strategies and coordination of renewable generation with conventional power sources [36, 53]. This sub-set forecasts integrative numerical weather prediction outputs and historical system behavior to anticipate broader weather-driven trends in solar generation [48, 51]. A smaller subset of forecasting tasks, such as long-term and seasonal horizons are associated with planning, performance assessment, and system design rather than real-time operation. These forecasts incorporate historical power output and climatological patterns [40, 62, 64] and estimate degradation effects in PV systems.

3.5 Taxonomy of machine learning models

The analysis of forecasting approaches presented in this section represents the methodological dimension of the literature synthesis, focusing on how different artificial intelligence paradigms are applied across forecasting horizons and application contexts.

Solar forecasting studies employing AI include a wide range of modeling paradigms that differ in learning strategy, data requirements, and application context (Fig. 4). Across the mapped literature, these approaches can be broadly categorized into classical ML models, deep learning architectures, ensemble and hybrid frameworks, transformer-based models, and emerging AI paradigms. Rather than focusing on algorithmic mechanics, the purpose of this classification is to highlight how different model families are applied across forecasting horizons, data environments, and operational scenarios. The reviewed evidence demonstrates that model selection is primarily driven by input

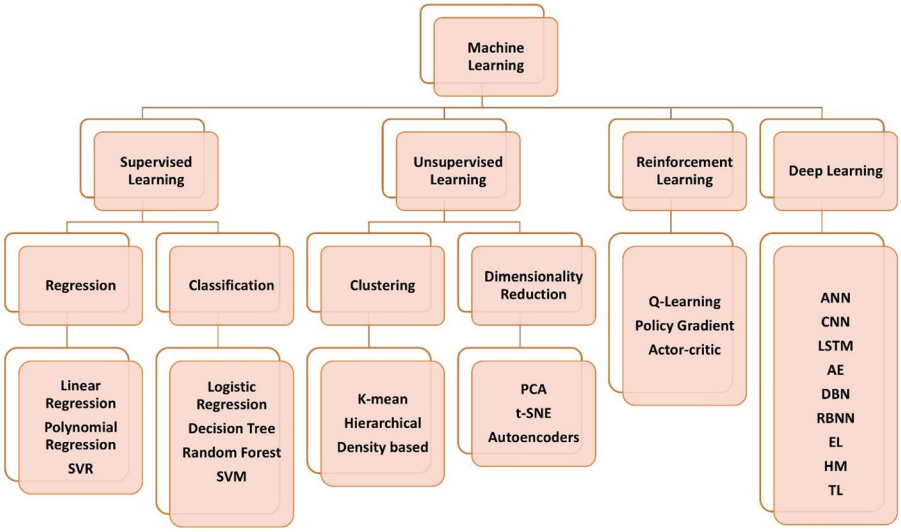


Fig. 4 Various Machine Learning methods used in solar energy forecasting. Support Vector Regression (SVR); Support Vector Machine (SVM); Principal Component Analysis (PCA); t-distributed Stochastic Neighbor Embedding (t-SNE); Artificial Neural Network (ANN); convolutional neural network (CNN); Long Short-Term Memory (LSTM); Auto Encoders (AE); Deep Belief Network (DBN); Radial Basis Neural Network (RBNN); Ensemble Learning (EL); Hybrid Model (HM); Transfer Learning (TL)

data characteristics, temporal structure, and the desired forecasting horizon, rather than purely by methodological novelty.

3.5.1 Classical machine learning models

Classical ML models remain widely used in solar forecasting applications, particularly in short-term and day-ahead prediction tasks, either as primary predictors or as benchmark references for evaluating more complex architectures [17, 18, 23, 24, 33, 34, 36, 38, 41, 42, 52, 53, 68, 69, 74]. Support vector regression (SVR) is repeatedly evaluated for both irradiance- and power-based forecasting with high accuracy in localized climatic settings. However, multi-site evaluations reveal that SVR exhibits reduced robustness across heterogeneous climates. Tree-based models such as Random Forest (RF) and Gradient Boosting (GB), emerge as the most consistently robust classical approaches. RF models achieve low error magnitudes in operational solar power forecasting. Across North African and Middle Eastern datasets, RF and GB models demonstrate R^2 values approaching 0.99. Such metrics reflect their capacity to capture nonlinear interactions among meteorological variables while maintaining resistance to overfitting. Linear regression (LR) and k-nearest neighbor (KNN) models are commonly included as baseline comparators. LR performs competitively under clear-sky and low-variability conditions, whereas KNN consistently exhibits higher error dispersion and sensitivity to noise. Figure 5 provides a schematic illustration of a support vector machine framework.

3.5.2 Deep learning models

Deep learning models constitute a foundational modeling layer across the mapped solar energy forecasting scenarios [22, 25, 43, 49, 51, 55, 65, 67]. Core deep learning models such as artificial neural networks (ANNs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs) are consistently represented across forecasting horizons, with model selection closely aligned to input structure and temporal scope. Across

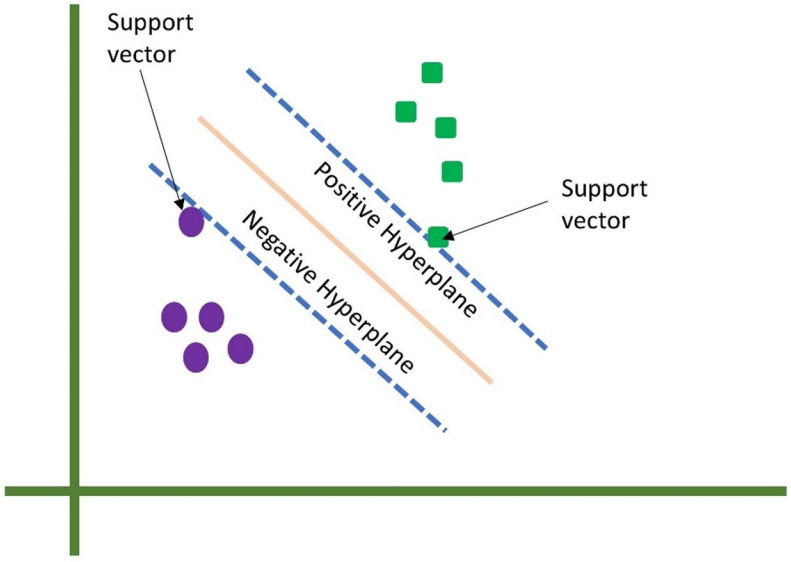


Fig. 5 Conceptual schematic of support vector machine decision boundaries illustrating separating hyperplanes and support vectors

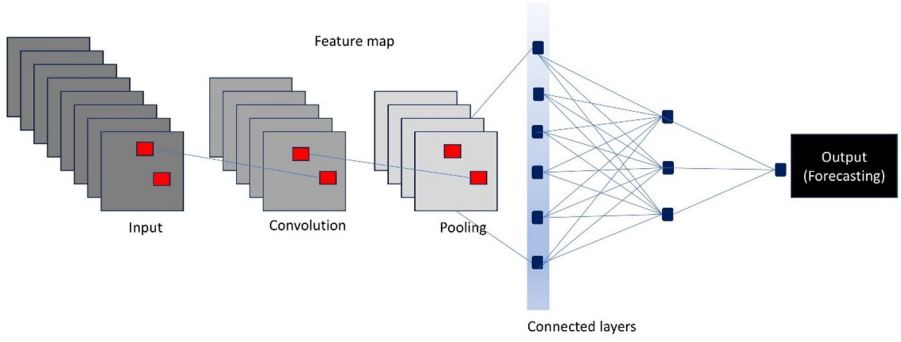


Fig. 6 Generalized architecture of a convolutional neural network for feature extraction and forecasting tasks

comparative evaluations, ANN models demonstrate strong nonlinear fitting capability and computational efficiency; however, their limited intrinsic temporal memory constrains performance under rapidly evolving atmospheric conditions and longer forecast horizons.

CNN models are reported primarily in studies incorporating spatially structured inputs, including satellite-derived irradiance fields, sky imagery, and gridded meteorological data (Fig. 6). Their application is concentrated in ultra-short-term and short-term forecasting. Evidence across the mapped literature indicates that CNNs excel at automated spatial feature extraction but exhibit reduced effectiveness when long-range temporal dependencies are required.

RNN-based models, on other hand represent the most widely adopted deep learning family for sequential solar irradiance and PV power time series. Benchmarking studies report that long short-term memory (LSTM) and gated recurrent units (GRU) models systematically outperform ANN and CNN baselines in temporally structured forecasting tasks with reported improvements in RMSE and mean absolute error (MAE) ranging from 15 to 40%, depending on climatic variability and input richness. However, studies

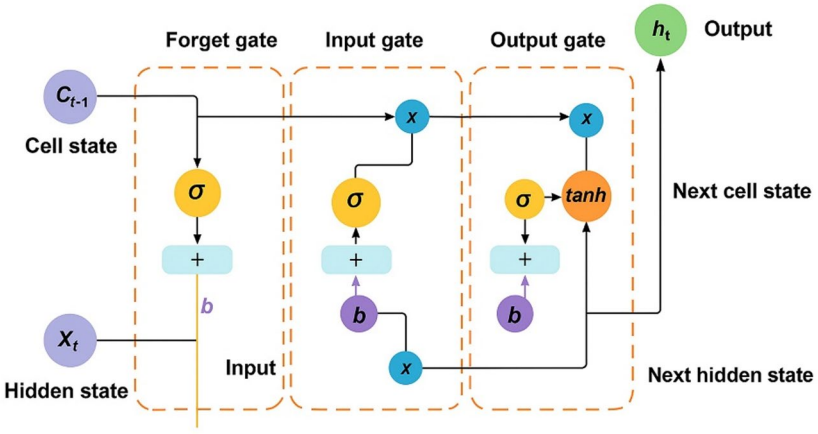


Fig. 7 Schematic representation of long short-term memory cell structure showing gating mechanisms and memory flow

document error accumulation and reduced stability under recursive multi-step prediction during abrupt cloud transitions. The mapped evidence places ANN, CNN, and RNN architectures as baseline deep learning approaches. Figure 7 depicts the internal gating and memory flow of LSTM architectures commonly used in solar forecasting.

3.5.3 Ensemble and hybrid models

Across the mapped literature, ensemble and hybrid modeling frameworks consistently emerge as structured responses to the intrinsic nonlinearity, nonstationarity, and stochastic variability of solar irradiance and PV power time series [21, 26, 30, 44, 59–61, 72]. Rather than representing isolated methodological choices, these approaches form a recurrent design pattern. A dominant trend identified in the evidence base is the use of competitive ensemble strategies, in which multiple base learners are trained independently and their outputs aggregated to stabilize prediction error. These ensembles frequently rely on data diversity and parameter diversity with tree-based models reported as effective base learners. Empirical evaluations summarized across studies demonstrate that such ensembles routinely outperform single learners, with reported reductions in RMSE ranging from approximately 4.6% to 14.6%, particularly for short-term and hourly forecasting tasks. In parallel, cooperative ensemble and hybrid frameworks are widely adopted to explicitly address signal irregularities prior to or following model training. Preprocessing-oriented hybrids including wavelet decomposition or empirical mode decomposition are used to decompose irradiance or PV output signals into frequency-specific components. Across multiple studies, these hybrids demonstrate improved temporal stability and reduced sensitivity to cloud-induced fluctuations, compared to non-decomposed baselines.

Post-processing hybrids represent a further recurrent configuration for operational forecasting contexts. Models combining linear time-series components with nonlinear are used to sequentially model structured trends and residual nonlinear dynamics. Evidence synthesized in the reviewed studies indicates that such post-processing ensembles

improve bias correction and short-horizon forecast reliability without requiring extensive historical datasets. The reviewed evidence indicates that these frameworks function as generalizable methodological solutions rather than location-specific optimizations. Their repeated appearance across forecasting targets and horizons signals a field-wide shift toward multi-stage and multi-learner architectures.

3.5.4 Transformer-based models

Transformer-based architectures constitute a recent but substantively evidenced methodological shift for multi-step, hourly, and day-ahead prediction tasks [31, 32, 73]. Across the mapped studies, these models are introduced explicitly to address limitations of RRNs, including error accumulation, vanishing gradients, and reduced stability under extended forecast horizons. Empirical evidence consistently demonstrates that attention-driven transformer models outperform recurrent baselines in long-horizon settings. In multi-step solar radiation forecasting, transformer models achieved 60% improvement in prediction accuracy while reducing degradation caused by recursive error propagation. Similar gains are reported in day-ahead PV power forecasting; yielded 48.3% lower mean absolute error compared with standalone LSTM approaches.

Prevalence of hybrid transformer–recurrent configurations rather than purely transformer-only models is concurrently observed in the evidence base. Studies integrating BiLSTM or GRU encoders with transformer attention and positional encoding consistently report near-perfect goodness-of-fit ($R^2 \sim 0.998$) alongside substantial reductions in RMSE and MAE across diverse climatic regions, including temperate and semi-humid environments. This hybridization reflects methodological convergence wherein recurrent components preserve short-term temporal memory, while transformers capture long-range dependencies and cross-variable interactions. Transformer-based models are further distinguished by their reliance on multivariate input streams and reduced dependence on locally measured irradiance. Although transformer-based approaches remain less prevalent than LSTM- or CNN-based models in absolute terms, their concentration in post-2022 literature, frequent association with day-ahead and multi-step horizons, and repeated demonstration of substantial error reductions indicate a positive methodological shift.

3.5.5 Emerging AI directions in solar energy forecasting

A set of emerging AI directions is observed that extends beyond established ML and deep learning frameworks [45–48, 50, 63, 64, 71]. These approaches appear primarily in response to recurring methodological challenges such as, scalability across sites, uncertainty representation, data governance constraints, and alignment with physical system behavior. Automated machine learning (AutoML) frameworks are reported in a limited but growing subset of studies, predominantly in contexts involving multi-site forecasting or operational deployment. Within the mapped evidence, AutoML is applied to automate model selection, feature selection, and hyperparameter optimization. These approaches are applied within forecasting pipelines where adaptability and reproducibility across varying datasets are prioritized, rather than as standalone predictive models.

Physics-informed ML emerges as another identifiable direction. These methods were observed in studies that integrate numerical weather prediction outputs or incorporate physical constraints related to system performance. Within the reviewed literature,

physics-informed approaches embed domain knowledge through constrained loss functions and hybrid physical–statistical representations. These methods are most frequently reported in hybrid modeling frameworks. Probabilistic and uncertainty-aware learning frameworks are increasingly reported, mostly in studies addressing short-term and day-ahead forecasting horizons. The mapped studies indicate the use of quantile regression, Bayesian neural networks, conformal prediction, and ensemble-based uncertainty estimation to characterize forecast uncertainty alongside point predictions. These approaches are primarily used in applications involving risk-sensitive analysis and market participation. Federated and distributed learning paradigms appear in a small but distinct subset of recent studies. These studies analyzed data privacy, ownership, and decentralization concerns inherent to modern energy systems. In certain instances, federated learning has been applied to provide collaborative model training across distributed PV installations without centralized data aggregation.

3.6 Data sources and feature engineering practices

Across the mapped evidence, ground-based measurements emerge as the most frequently utilized data source for both solar irradiance and PV power forecasting. These datasets consist of time-series observations of global or component irradiance, ambient temperature, relative humidity, wind speed, and historical PV power output [32, 38, 45, 49], Studies addressing spatially distributed solar resource assessment, utilized satellite-derived data as a complementary input source. Remote sensing data providing information on cloud cover, aerosol loading, surface albedo, and atmospheric motion are commonly integrated to extend spatial coverage beyond dense ground-based sensor networks [20, 51, 59]. The use of satellite inputs is most frequently associated with short-term and ultra-short-term forecasting horizons. Likewise, NWP outputs are frequently incorporated in short-term and day-ahead forecasting contexts. The mapped literature indicates that NWP data are primarily used to provide forward-looking meteorological variables which are combined with historical observations to capture both persistence effects and forecast-driven trends [52, 53]. These practices are prevalent in studies oriented toward grid scheduling. In addition to single-source datasets, hybrid data configurations are widely reported. Hybrid datasets integrate ground-based measurements, satellite observations, and NWP outputs to enhance robustness and spatial representativeness. Such integrations are most commonly observed in multi-site forecasting studies, data-scarce environments, and regions characterized by complex or highly variable climatic conditions [31, 48, 54].

Data preprocessing and feature engineering practices are consistently described as integral components of forecasting pipelines. The mapped studies frequently report preprocessing steps including data cleaning, handling of missing or inconsistent values, normalization or scaling of input variables, and temporal alignment across datasets with differing resolutions [56, 57]. These practices are required to ensure consistency when using heterogeneous data sources. Feature engineering strategies reported across the literature emphasize temporal representation. Commonly constructed features include hour-of-day, day-of-year, and seasonal indicators, which are used to encode diurnal and seasonal patterns in solar availability [65]. Lagged variables derived from historical irradiance or PV power output are also frequently reported, particularly in short-term forecasting tasks, to capture temporal dependencies and persistence effects [44, 64]. To

address high-dimensional input spaces and multicollinearity, dimensionality reduction and feature selection techniques are routinely applied. Principal component analysis and supervised feature selection methods are reported to reduce input complexity while retaining salient information relevant to forecasting performance [39, 69, 70]. In image-based forecasting applications, learned feature representations extracted through deep learning architectures serve an analogous role by compressing high-dimensional spatial inputs into informative latent features [46, 51].

3.7 Performance metrics reported

The evaluation of solar irradiance and PV power forecasting models report a diverse set of quantitative performance metrics that detect prediction accuracy and error magnitude. Deterministic error metrics form the dominant basis for model assessment across forecasting targets. MAE and RMSE are consistently reported to quantify deviations between predicted and observed values of solar irradiance or PV power output (Eqs. 1 and 2) [60, 61]. These metrics are applied across ultra-short-term, short-term, and day-ahead forecasting tasks and are used in both irradiance-based and power-based prediction frameworks. The coefficient of determination (R^2) is also frequently reported as an indicator of goodness-of-fit between modeled and observed outputs, particularly within regression-oriented formulations (Eq. 3) [59].

$$\mathrm{MAE} = \frac{1}{\mathrm{n}} \sum_{i=1}^{n} \mathrm{y_i} - \acute{\mathrm{Y}}_{\mathrm{i}} \tag{1}$$

where yi is the actual value and Ýi is the predicted value.

$$\mathrm{RMSE} = \sqrt{\frac{1}{\mathrm{n}} \sum_{i=1}^{n} (\mathrm{y_i} - \acute{\mathrm{Y}}_{\mathrm{i}})^2} \tag{2}$$

$$\mathrm{R}^2 = 1 - \frac{\sum_{i=1}^{n} (\mathrm{yi} - \widehat{\mathrm{y}}\mathrm{i})^2}{\sum_{i=1}^{n} \left(\mathrm{yi} - \bar{\mathrm{y}}\right)^2} \tag{3}$$

In addition to absolute error measures, relative and normalized metrics are reported to facilitate comparison across sites, datasets, and system capacities. Metrics such as, forecast skill score (Eq. 4), normalized RMSE and percentage-based errors are used to account for differences in irradiance magnitude, installed capacity, or climatic context, and are especially relevant in multi-site and regional-scale forecasting studies [33, 49, 54].

$$\text{Skill score} = \frac{\mathrm{RMSE}_{\mathrm{model}} - \mathrm{RMSE}_{\mathrm{reference}}}{\mathrm{RMSE}_{\mathrm{model}}} \tag{4}$$

$\mathrm{RMSE}_{\mathrm{model}}$ is the root mean square error of the forecasting model, and $\mathrm{RMSE}_{\mathrm{reference}}$ is the root mean square error of the baseline model.

A subset of studies reports skill-based or comparative metrics to contextualize forecasting performance relative to baseline or reference models. Forecast skill scores are used to express improvement over persistence models or simpler statistical benchmarks,

particularly in short-term operational applications where incremental performance gains are operationally meaningful [62]. In forecasting frameworks that incorporate uncertainty quantification, probabilistic performance metrics are also reported. These metrics assess prediction interval coverage, reliability, or distributional consistency and are primarily associated with probabilistic and uncertainty-aware forecasting approaches [21, 51]. Such metrics are most commonly reported in short-term and day-ahead forecasting contexts, where uncertainty characterization supports risk-informed grid operation and market participation.

4 Discussion

The discussion integrates understandings from the methodological and data-oriented analyses to highlight operational implications, deployment challenges, and research gaps associated with AI-based solar forecasting systems. In addition to methodological understandings, these findings provide important implications for practical deployment of forecasting systems within modern power grids and renewable energy management infrastructures.

The methodological patterns observed across recent solar irradiance and PV power forecasting data indicate a clear alignment between model choice and the inherent characteristics of solar energy data [77]. The increasing dominance of deep learning and ensemble-based approaches emphasizes their suitability for determining complex interactions among meteorological, atmospheric, and system-level variables [78]. While recurrent neural networks and LSTM architectures are particularly well matched to time-series forecasting tasks, convolutional neural networks are naturally aligned with image-based inputs [79]. Owing to the increasing availability of high-resolution temporal and spatial data, deep learning models are taking gaining lead. Evidences suggest that architectures such as ANN, CNN, and LSTM are consistently selected based on the structure of input data rather than novelty [80–82]. CNNs employment are favoured for spatially heterogeneous cloud fields whereas LSTM-based models dominate temporally persistent forecasting tasks [83]. However, the mapped evidence also highlights systematic limitations of standalone deep learning models, such as diminishing sensitivity to weather regime shifts and error accumulation in extended horizons [84]. Hybrid and ensemble models emerge as a methodological response to these limitations, rather than a distinct paradigm shift. Their prevalence across multiple forecasting horizons suggests that combining complementary inductive biases such as, decomposition with learning, or stacking heterogeneous learners, has become a pragmatic strategy to stabilize predictions under nonstationary atmospheric conditions [6, 85–87].

Forecast horizon and sky-condition variability are primary drivers of model selection. Ultra-short-term forecasting is dominated by spatially aware models and tree-based learners [88]. Short-term and day-ahead forecasting increasingly rely on recurrent and attention-based architectures capable of capturing temporal persistence and lag effects [89]. Importantly, model performance is consistently climate-dependent. Classical and linear models perform competitively under clear-sky or weakly variable conditions, while their accuracy degrades during rapidly evolving cloud transitions [90]. Conversely, CNN-based and hybrid models demonstrate their strongest advantages under partially cloudy regimes [91]. These observations highlight that no single model class can be considered universally optimal across all meteorological regimes.

Despite methodological advances, the review identifies persistent data-related constraints. A substantial proportion of studies rely on single-site or region-specific datasets. While high accuracy is frequently reported, the lack of cross-site validation raises concerns regarding model transferability. Several studies attempt to address data scarcity through feature engineering, dimensionality reduction, or augmentation strategies; however, standardized data practices remain limited. The growing use of external numerical weather prediction outputs and satellite-derived inputs echoes an emerging effort to reduce dependence on local measurements, yet their integration remains inconsistent across the literature [92]. These challenges become particularly significant when forecasting systems are deployed across geographically diverse regions. Evidence synthesized suggests that models optimized under specific environmental conditions may experience performance degradation when transferred to different locations without adaptive recalibration. Consequently, operational deployment requires forecasting frameworks capable of incorporating regional climate characteristics. Emerging approaches provide pathways for improving generalization across heterogeneous geographic environments. Furthermore, in regions with highly variable weather patterns, uncertainty-aware forecasting becomes essential for reliable grid operation. Probabilistic, ensemble, and Bayesian learning frameworks provide confidence bounds and risk-sensitive predictions. Expanding the integration of such uncertainty quantification mechanisms into deployment-oriented forecasting pipelines therefore represents a research direction for climate-resilient energy systems.

Forecasting accuracy carries significant economic implications across renewable energy systems. The improvements reported in the reviewed literature, such as reductions in RMSE of approximately 4.6% to 14.6% for ensemble frameworks and substantial decreases in mean absolute error for transformer-based architectures, indicate potential financial benefits. In electricity markets with high solar penetration, forecast deviations directly translate into reserve curtailment losses and market settlement penalties; therefore, even modest improvements in prediction accuracy yields measurable economic value for system operators and energy producers [93]. Improved short-term and day-ahead forecasts also support optimized unit commitment and dispatch decisions.

From an investment and planning perspective, reliable forecasting reduces uncertainty associated with photovoltaic project performance, which is a critical factor in financial modeling and infrastructure planning. A representative operational example demonstrates the economic value of AI-based probabilistic forecasting integrated with stochastic market optimization [45]. In a grid-connected photovoltaic system participating in electricity markets, uncertainty-aware forecasting improved bidding strategies, resulting in approximately 22% higher revenues and nearly 50% reduction in imbalance costs compared with conventional approaches. This case illustrates how forecasting accuracy translates into tangible financial and operational benefits when coupled with decision-optimization frameworks [45].

Accurate generation prediction permits more effective storage sizing and capacity expansion decisions, thereby improving return-on-investment predictability. In emerging energy systems, AI-driven forecasting tools are increasingly viewed as enabling technologies. Furthermore, the integration of forecasting platforms with digital grid management systems creates opportunities for value-added services such as predictive

maintenance scheduling, automated energy trading strategies, and intelligent demand–supply coordination.

The mapped evidence highlights that forecasting accuracy is a means to operational decision-making. Improved short-term and day-ahead forecasts are directly linked to enhanced grid stability and reduced reserve requirements [94]. Transformer-based and probabilistic forecasting approaches, although less prevalent, are increasingly aligned with market participation and risk-aware planning [95]. However, operational deployment considerations such as, retraining frequency, explainability, and robustness under unseen conditions, are rarely evaluated systematically. Recent advancements in lightweight and explainable AI frameworks for distributed sensing environments further emphasize the importance of interpretable and computationally efficient models for real-world energy applications [96]. These observations also echoes growing industrial demand for reliable forecasting tools capable of supporting grid operators, energy traders, and infrastructure planners in environments with high renewable penetration and operational uncertainty.

Several research gaps emerge from this scoping review. First, there is a need for systematic cross-climate and cross-site benchmarking, moving beyond isolated case studies. Second, uncertainty quantification remains underexplored relative to deterministic accuracy, despite its importance for operational risk management. Third, while transformer-based and emerging AI approaches show promise, their data requirements and deployment feasibility require further investigation. Future research would benefit from integrating physics-informed constraints, probabilistic frameworks, and federated learning strategies, particularly to enhance generalization across geographically distributed PV systems. Additionally, greater emphasis on reproducibility, open datasets, and standardized evaluation protocols would significantly strengthen the evidentiary foundation of AI-based solar forecasting research. It is important to note that the trends identified in this review are derived from publication patterns within the academic literature and may not directly reflect industry adoption or commercial deployment levels, as technological translation into operational energy systems often occurs with temporal lag and additional practical constraints.

5 Future outlook

The evolving role of AI–driven solar forecasting is expected to extend beyond predictive accuracy toward integrated energy system optimization and cross-sectoral decarbonization. Future research should prioritize the development of transferable and uncertainty-aware forecasting frameworks that can support infrastructure planning and operational decision-making in increasingly complex energy ecosystems. For instance, accurate renewable generation prediction will be critical for optimizing the spatial deployment and operational scheduling of green hydrogen electrolysers, where forecasting uncertainty directly influences transmission congestion management and investment efficiency [97]. In parallel, solar forecasting is likely to play an important role in the decarbonization of urban heat sectors through coordinated building energy management [98]. Emerging methodological directions such as federated learning, physics-informed modeling, and probabilistic forecasting may further improve robustness and scalability across geographically distributed energy networks. Additionally, future studies should emphasize explainability, adaptive retraining strategies, and real-time

deployment considerations to bridge the gap between algorithmic advancements and operational implementation. These developments suggest that AI-based solar forecasting will play an increasingly central role in enabling resilient, decentralized, and low-carbon energy systems.

6 Conclusion

This scoping review systematically mapped the literature on AI-based solar energy forecasting across diverse forecasting contexts. There is a clear methodological progression from classical ML approaches toward deep learning, hybrid ensemble models, transformer-based architectures, and emerging artificial intelligence paradigms. This shows increasing data availability and computational capability within the field. Short-term and day-ahead forecasting horizons dominate the research landscape, while comparatively fewer studies address long-term and seasonal forecasting contexts. The majority of studies rely on site-specific datasets with limited cross-regional validation. Although deterministic accuracy metrics are widely reported, uncertainty-aware forecasting approaches and probabilistic evaluation frameworks remain relatively underrepresented despite their importance for operational risk management. The review finds a clear transition from conventional AI solar to emerging paradigms forecasting. Thus, providing a strong potential to address scalability and uncertainty challenges. This review underscores the need for more transferable, uncertainty-aware, and operationally grounded AI frameworks for solar forecasting.

Abbreviations

PV	Photovoltaic
AI	Artificial intelligence
ML	Machine learning
PRISMA-ScR	Preferred reporting items for systematic reviews and meta-analyses extension for scoping reviews
NWP	Numerical weather prediction
SVM	Support vector machine
DT	Decision trees
RF	Random forests
ANN	Artificial neural networks
CNN	Convolutional neural networks
RNN	Recurrent neural networks
LSTM	Long short-term memory
GRU	Gated recurrent units
RMSE	Root mean square error
MAE	Mean absolute error
AutoML	Automated machine learning

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Grammarly was used for language improvement. The content generated by the AI tool was reviewed/edited and we are responsible for all the content in our manuscript.

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